PHASE 0
Literature Retrieval

Input: User Problem

e.g., How to improve the sample efficiency of RL in sparse reward settings?

Multi-Source Retrieval

DBLP / OpenAlex / OpenReview / arXiv

- Query expansion
- Time window filtering
- Deduplication
- Quality filtering

Output
30–40 Relevant Papers

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PHASE 1
Problem Diagnosis Literature-Grounded

1.1 Gap Extraction

- Summarize the state of the art based on retrieved papers
- Identify limitations and unresolved gaps
- Ground the gap in specific evidence (≥ 2 papers)

1.2 Assumption Analysis

- Identify critical assumptions in prior work
- Analyze whether they are overly strong or limit generalization
- Surface overlooked factors

Output: Target Bottleneck

A precise, actionable bottleneck with citations (≥ 2 papers) and justification.

Proceed

Do Not Generate (Stop)

PHASE 2
Pattern-Guided Idea Synthesis

2.1 Pattern Selection

- Select 1–3 high-level innovation patterns from the pattenn library based on structural fit to the bottleneck (not frequency).

Audit Assumption | Alternative Formalism | Latent Alignment

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2.2 Idea Instantiation

Instantiate a concrete idea using the selected patterns.

Idea Candidate (Minimal Signature)

Core Idea: Learn a conservative latent dynamics model with uncertainty-aware planning to reduce extrapolation error in offline RL.

Signature Terms: (latent dynamics, uncertainty, conservative, OOD, offline RL)

What to Run: D4RL (medium-replay) + AntMaze

Compute Budget: ~4 × A100 (3 days)

Mechanism Probe: (1) OOD value gap
(2) Uncertainty calibration
(3) Planning decision shift

PHASE 3
Corpus-Grounded Adversarial Audit

Challenge the idea against real literature

3.1 Novelty Check (Collision)

Retrieve recent (≤ 6 months) papers using the signature terms and detect potential collisions.

Output: Collision Hits (Top ~k papers)

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3.2 Feasibility Check (Anti-pattern)

Evaluate risks of "paper without substance", e.g., weak novelty, no mechanism, unverifiable claims, or overclaimed contributions.

3.3 Coverage Check (Paper-level Threat)

Check if any existing paper (from Phase 0 and collision hits) substantially covers or renders the idea redundant.

Audit Decision

Advance (To Phase 4) | Revise (Loop Back) | Abandon (Stop)

PHASE 4
Structured Idea Formulation (Idea Card)

4.1 Motivation & Problem
Why prior work fails and why this idea can succeed.

4.2 Method Sketch
High-level approach and key components.

4.3 Testable Hypothesis
Clear, falsifiable claims and expected gains.

4.4 Mechanism Probe
How to verify the underlying mechanism.

4.5 Evaluation Plan
Datasets, metrics, baselines, and ablation plan.

4.6 Feasibility Analysis (5D)
Technical / Data / Compute / Novelty / Impact.

Output: Idea Card

Innovation Pattern Library (Knowledge Backbone)

13 High-Level Patterns | 47 Sub-patterns | Methodology Cards (Successful Execution, Reject Failure Modes, Reviewer Focus, Exemplars)

Used in: Phase 2 (Synthesis)
Phase 3 (Audit Reference)

Cross-Cutting Knowledge & Resources

- Paper Corpus (ICLR/ICML/NeurIPS 2021–2025) N = 1,947
- Innovation Pattern Library 13 High-level Patterns 47 Sub-patterns
- Methodology Cards (Successful Execution, Reject Failure Modes, Reviewer Focus, Exemplars)
- Domain Pattern Distributions (28 Domains)
- Statistics & Priors (Pattern Composition, Acceptance Rates, Review vs. Citation)

System Guarantees (Invariants)

1 Kill-switch Integrity

The following fields remain byte-identical from Phase 2 → Phase 4.

- What to Run
- Compute Budget
- Mechanism Probe

2 Expansion Completeness

The final idea card must include all required sections in Phase 4.

Legend

- Input / Output
- LLM / Reasoning
- Database / Retrieval
- Audit / Decision
- Loop / Iteration
- Artifact